Rainul Hakim

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Education

Davidson College — Davidson, NC
B.S. Computer Science, Minor in Data Science | GPA: 3.757 — *Expected May 2029*
- Relevant Coursework: Data Structures, Database Programming, Linear Algebra, Calculus I & II

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, SQL, C, R, HTML/CSS
Web & Backend: React, Next.js, Node.js, Tailwind CSS, REST APIs, Zod, Serverless Functions, Web Push
Cloud & Databases: PostgreSQL, Supabase, AWS (Lambda, RDS, API Gateway, EventBridge), Vercel
ML & Data: PyTorch, PyTorch Geometric, TensorFlow.js, scikit-learn, OpenCV, MediaPipe, NumPy, Pandas
Tools: Git, GitHub, Linux, VS Code, Jupyter, HPC Clusters, Mapbox, Claude Code, Codex

Projects

Left No Crumbs! – 1st Place ($1,000), IdeaSprint | ***Next.js, Supabase, PostgreSQL, Web Push***
- Co-founded and built a real-time food-rescue web app alerting **600**+ students across **55** organizations to surplus catered meals
- Designed a **PostgreSQL** schema with row-level security and campus-only auth; added dietary filters and rate-limited alerts
- Partnered with Davidson Dining and Technology & Innovation on an integration with the Davidson One campus app

CatCruxis – Campus Event Platform | ***Python, PostgreSQL, AWS, React, Mapbox***
- Designed a normalized (3NF) **PostgreSQL** schema unifying events from WildcatSync, Handshake, and other campus sources
- Automated an ETL pipeline with **EventBridge**-scheduled **AWS Lambda** jobs to extract, clean, and load event data into **RDS**
- Built a **React**/**Mapbox** map interface to filter events by location, date, and category

AI Meal Analysis Platform | ***React, Vite, JavaScript, Gemini API, Vercel***
- Shipped a **React** web app used by **100**+ students that turns campus meals into macro breakdowns, health flags, and alternatives
- Secured the **Gemini API** behind **serverless functions** with JSON validation; built meal comparison and weekly scores

In-Browser Hand Gesture Recognition | ***JavaScript, TensorFlow.js, MediaPipe***
- Ran real-time (∼**30 fps**) gesture classification fully in-browser with **MediaPipe** landmarks and a **TensorFlow.js** model
- Added an in-browser training UI and hand-size-invariant landmark normalization; rendered segmentation effects on Canvas

Experience

Software Engineering Intern — Aug 2026 – Present
StudyBuddy — *Davidson, NC*
- Built an AI study-planning app with **Next.js**, **React** and **TypeScript** API routes, piloted with 30 Davidson students
- Integrated the **Claude API** with automatic **Gemini API** fallback, keeping AI features up during provider outages
- Made AI output reliable with **Zod**-validated response schemas and a ∼15-case **Node.js** guardrail test suite

Tech Consultant — Aug 2026 – Present
DataCATS (Consulting, Analytics, Tutoring Services), Davidson College — *Davidson, NC*
- Led **85**+ sessions and projects for students and faculty, debugging code and data pipelines and guiding modeling

Machine Learning Research Intern — May 2026 – Aug 2026
Facility for Rare Isotope Beams (FRIB) — *East Lansing, MI*
- Developed a self-supervised **DGCNN** encoder (**PyTorch Geometric**) for **AT-TPC** point-cloud data, toward a reusable **foundation model** for FRIB experiments
- Used **SimCLR**-style contrastive learning on **60,795** unlabeled events to reach **88.8%** accuracy / **0.886** weighted F1 on 3-class particle-track classification with only ∼**2.5K** labels
- Ran **14**+ versioned experiments on an **HPC** cluster; a physics-based z-rotation augmentation gave the largest gain
- Debugged the point-cloud data pipeline (feature-dimension bug, sentinel-value handling) and shared documented code via **Git** in a cross-institutional research codebase

Leadership & Involvement

Logistics, Hack@Davidson — 2026 – Present
Secretary, Association for Computing Machinery (ACM), Davidson College — 2026 – Present
Resident Advisor, Davidson College — 2026 – Present
Event Coordinator, Davidson Entrepreneurship Club — 2025 – Present
Treasurer, Davidson International Association — 2026 – Present